Assignment 1 - Airbnb Price Prediction pipeline

Student Details:-

- Name: Praveen Kumar Saravanan
- Student Number: S4163448
- Course Code: COSC2793

I built and evaluated a supervised machine learning pipeline to predict Airbnb nightly price using listing, host, and location features. This notebook uses a fixed seed of 42 (`RANDOM_STATE = 42`) to ensure reproducible results.

`sns.set_theme(style='whitegrid')` is used for consistent chart styling and readable plots.

How to run this notebook

1. Keep `train_data.csv` and `test_data.csv` in the same folder as this notebook.
2. Run all cells from top to bottom.
3. The predictions file is saved as `s4163448_predictions.csv` based on the value set in the imports cell.
4. Required packages: numpy, pandas, seaborn, matplotlib, scikit-learn.

In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.model_selection import train_test_split, KFold, cross_validate
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

RANDOM_STATE = 42
student_number = 's4163448'
sns.set_theme(style='whitegrid')
```

Task Definition and Evaluation Framework

- **Task type**: Supervised regression.
- **Target variable**: `price` (nightly listing price).
- **Why regression**: `price` is continuous and should be estimated from available features.

Evaluation Design

- Hold-out split: 80% training, 20% validation.
- Stability estimate: 5-fold cross-validation on training split.
- Metrics:
  - MAE: average absolute error in price units (easy business interpretation).
  - RMSE: penalizes large errors more strongly than MAE.
  - R2: proportion of variance explained by the model.

Dataset overview

- Train data contains the target column `price`; test data does not.
- Features include host status, location, room type, capacity, rooms, review metrics, and host listings count.
- Data types and descriptive statistics are shown in the tables below.
- Missing values are checked and summarized in the next section.

In [2]:
```
train_df = pd.read_csv('train_data.csv')
test_df = pd.read_csv('test_data.csv')

print('Train shape:', train_df.shape)
print('Test shape:', test_df.shape)
display(train_df.head())
display(train_df.dtypes.to_frame('dtype'))
```

```
Train shape: (8586, 16)
Test shape: (8585, 15)
```

	host_is_superhost	city	country	latitude	longitude	room_type	accommod
0	f	Kingston	Australia	-38.022565	145.098937	Entire home/apt	
1	f	Yarra	Australia	-37.782125	144.988079	Entire home/apt	
2	t	Port Phillip	Australia	-37.853710	144.981529	Entire home/apt	
3	f	Glen Eira	Australia	-37.876393	145.036635	Entire home/apt	
4	f	Melbourne	Australia	-37.810398	144.950113	Entire home/apt	

	dtype
host_is_superhost	str
city	str
country	str
latitude	float64
longitude	float64
room_type	str
accommodates	int64
bathrooms	float64
bedrooms	float64
beds	float64
price	int64
minimum_nights	int64
number_of_reviews	int64
review_scores_rating	float64
instant_bookable	str
calculated_host_listings_count	int64

In [3]:
```
display(train_df.describe(include='all').T)
print('Missing values (train):')
display(train_df.isna().sum()[train_df.isna().sum() > 0])
```

	count	unique	top	freq	mean	std	
host_is_superhost	8586	2	f	6105	NaN	NaN	
city	8586	30	Melbourne	2917	NaN	NaN	
country	8586	1	Australia	8586	NaN	NaN	
latitude	8586.0	NaN	NaN	NaN	-37.825074	0.065145	-
longitude	8586.0	NaN	NaN	NaN	145.009374	0.130473	1
room_type	8586	3	Entire home/apt	5771	NaN	NaN	
accommodates	8586.0	NaN	NaN	NaN	3.559865	2.225801	
bathrooms	8586.0	NaN	NaN	NaN	1.28232	0.557906	
bedrooms	8586.0	NaN	NaN	NaN	1.533077	0.916369	
beds	8586.0	NaN	NaN	NaN	2.036222	1.544967	
price	8586.0	NaN	NaN	NaN	139.771372	123.293376	
minimum_nights	8586.0	NaN	NaN	NaN	3.158747	20.788007	
number_of_reviews	8586.0	NaN	NaN	NaN	27.946657	42.409812	
review_scores_rating	8586.0	NaN	NaN	NaN	94.176101	8.471527	
instant_bookable	8586	2	f	4461	NaN	NaN	
calculated_host_listings_count	8586.0	NaN	NaN	NaN	7.427324	15.438001	

```
Missing values (train):
Series([], dtype: int64)
```

Per-feature statistics and categorical distributions

Numeric summary includes all numeric columns (including `price`), and categorical summary lists unique values plus the most frequent category and its count.

In [4]:
```
# Per-feature statistics
numeric_stats = (
    train_df.select_dtypes(include=[np.number])
    .agg(['min', 'max', 'mean', 'median', 'std'])
    .T
    .round(3)
)

display(numeric_stats)

cat_cols = train_df.select_dtypes(include=['object', 'string', 'category']).colu
if len(cat_cols) > 0:
    categorical_stats = train_df[cat_cols].describe().T[['unique', 'top', 'freq'
    display(categorical_stats)
else:
    print('No categorical features found.')
```

	min	max	mean	median	std
latitude	-38.224	-37.491	-37.825	-37.816	0.065
longitude	144.532	145.760	145.009	144.977	0.130
accommodates	1.000	16.000	3.560	3.000	2.226
bathrooms	0.000	9.000	1.282	1.000	0.558
bedrooms	0.000	10.000	1.533	1.000	0.916
beds	0.000	18.000	2.036	2.000	1.545
price	0.000	3000.000	139.771	115.000	123.293
minimum_nights	1.000	1000.000	3.159	2.000	20.788
number_of_reviews	1.000	479.000	27.947	11.000	42.410
review_scores_rating	20.000	100.000	94.176	97.000	8.472
calculated_host_listings_count	1.000	98.000	7.427	1.000	15.438

	unique	top	freq
host_is_superhost	2	f	6105
city	30	Melbourne	2917
country	1	Australia	8586
room_type	3	Entire home/apt	5771
instant_bookable	2	f	4461

In [5]:
```
# Missingness profile
missing_counts = train_df.isna().sum().sort_values(ascending=False)
missing_counts = missing_counts[missing_counts > 0]

if len(missing_counts) > 0:
    plt.figure(figsize=(8, 4))
    sns.barplot(x=missing_counts.index, y=missing_counts.values, color='steelblu
    plt.xticks(rotation=45, ha='right')
    plt.title('Missing values per feature (train)')
    plt.ylabel('Count of missing values')
    plt.tight_layout()
    plt.show()
else:
    print('No missing values in training data.')
```

```
No missing values in training data.
```

Initial Data Interpretation

The dataset contains **8,586 training rows** and **8,585 test rows**. The training data has 16 columns (15 predictors + 1 target `price`), while the test data has 15 predictor columns.

Feature types

- Categorical: `host_is_superhost`, `city`, `country`, `room_type`, `instant_bookable`
- Numerical: `latitude`, `longitude`, `accommodates`, `bathrooms`, `bedrooms`, `beds`, `minimum_nights`, `number_of_reviews`, `review_scores_rating`, `calculated_host_listings_count`

Missingness

No missing values were detected in the provided training set at runtime. Imputation remains in the pipeline as a robust safeguard for reproducibility and for potential future data with incomplete records.

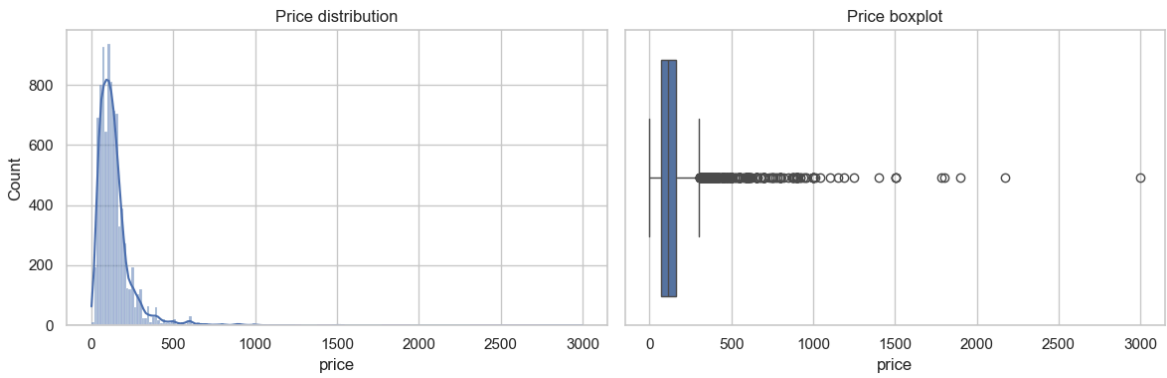
Validity note

Even with complete data, learned relationships remain correlational. Pricing can also be influenced by unobserved factors (seasonality, event periods, listing quality details), which limits causal interpretation.

In [6]:
```
X = train_df.drop(columns=['price'])
y = train_df['price']

categorical_features = X.select_dtypes(include=['object', 'string', 'category'])
numeric_features = [c for c in X.columns if c not in categorical_features]

print('Categorical:', categorical_features)
print('Numeric:', numeric_features)
```

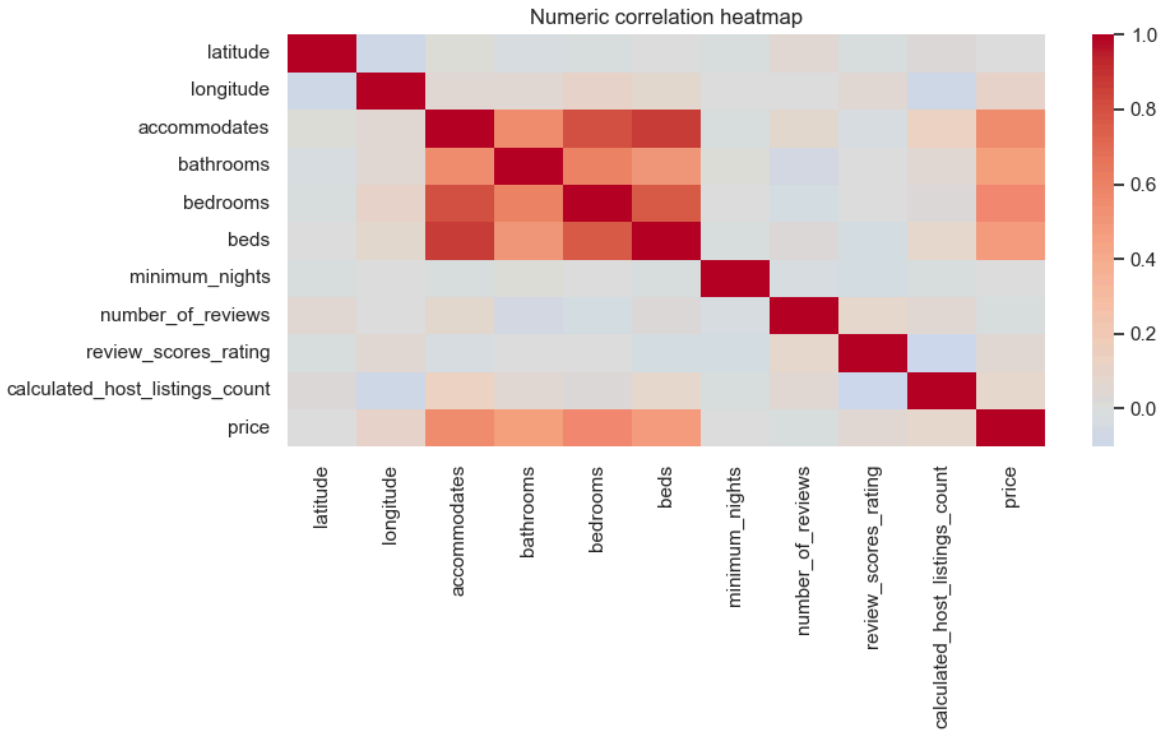
```
Categorical: ['host_is_superhost', 'city', 'country', 'room_type', 'instant_booka
ble']
Numeric: ['latitude', 'longitude', 'accommodates', 'bathrooms', 'bedrooms', 'bed
s', 'minimum_nights', 'number_of_reviews', 'review_scores_rating', 'calculated_ho
st_listings_count']
```

In [7]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
sns.histplot(y, kde=True, ax=axes[0])
axes[0].set_title('Price distribution')
sns.boxplot(x=y, ax=axes[1])
axes[1].set_title('Price boxplot')
plt.tight_layout()
plt.show()
```



In [8]:
```
corr = train_df[numeric_features + ['price']].corr(numeric_only=True)
plt.figure(figsize=(10, 6))
```

```
sns.heatmap(corr, cmap='coolwarm', center=0)
plt.title('Numeric correlation heatmap')
plt.tight_layout()
plt.show()
```



EDA Interpretation

The `price` distribution is right-skewed with visible high-price outliers, indicating that a minority of listings are substantially more expensive than typical listings. This motivates using both MAE and RMSE: MAE for average error magnitude, RMSE for sensitivity to large misses.

From the correlation heatmap, location and capacity-related variables (for example `accommodates`, `bedrooms`, and geographic coordinates) show meaningful linear relationships with `price`, though strengths vary and are not sufficient alone for perfect prediction.

All available features are retained initially to avoid premature feature dropping. Regularized models (Ridge/Lasso) then provide controlled complexity and help reduce the impact of weak or redundant predictors.

Correlation results are used for interpretation and model design only; they do not establish causation.

In [9]:
```
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_st

preprocessor = ColumnTransformer(
    transformers=[
        ('num', Pipeline([('imputer', SimpleImputer(strategy='median')), ('scale
        ('cat', Pipeline([('imputer', SimpleImputer(strategy='most_frequent')),
    ]
)
```

```
kfold = KFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)

def metrics(y_true, y_pred):
    return {
        'MAE': mean_absolute_error(y_true, y_pred),
        'RMSE': np.sqrt(mean_squared_error(y_true, y_pred)),
        'R2': r2_score(y_true, y_pred)
    }

def evaluate(name, model):
    pipe = Pipeline([('preprocess', preprocessor), ('model', model)])
    pipe.fit(X_train, y_train)

    val_pred = pipe.predict(X_val)
    val_m = metrics(y_val, val_pred)

    cv = cross_validate(
        pipe, X_train, y_train, cv=kfold,
        scoring=('neg_mean_absolute_error', 'neg_root_mean_squared_error', 'r2')
        n_jobs=-1
    )

    return {
        'Model': name,
        'Val_MAE': val_m['MAE'],
        'Val_RMSE': val_m['RMSE'],
        'Val_R2': val_m['R2'],
        'CV_MAE': -cv['test_neg_mean_absolute_error'].mean(),
        'CV_RMSE': -cv['test_neg_root_mean_squared_error'].mean(),
        'CV_R2': cv['test_r2'].mean()
    }, pipe
```

Pipeline Design

- `ColumnTransformer` ensures each feature type gets appropriate preprocessing.
- Median imputation is robust for skewed numerical features.
- Most-frequent categorical imputation is simple and consistent for low missingness.
- One-hot encoding makes categorical values usable by linear models.
- Standardization is important for fair regularization in Ridge/Lasso.
- Wrapping preprocessing + model in a single pipeline reduces leakage risk.

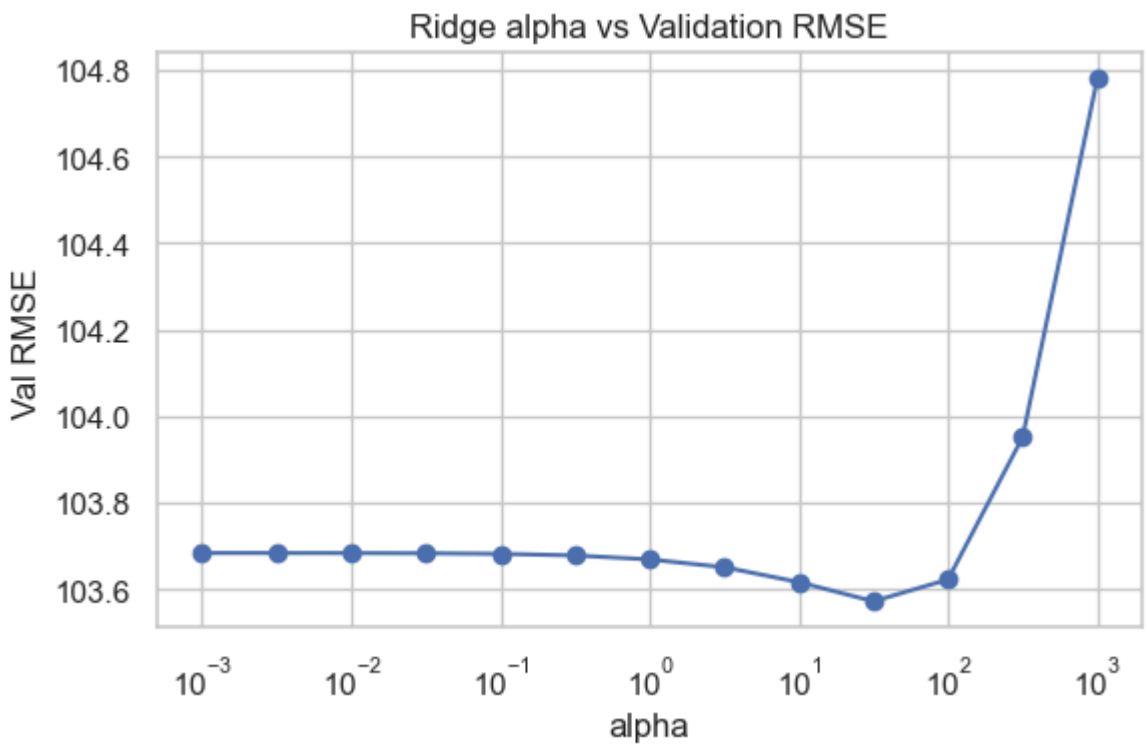
In [10]:
```
# Model 1: Linear Regression
linear_row, linear_pipe = evaluate('LinearRegression', LinearRegression())
display(pd.DataFrame([linear_row]))
```

	Model	Val_MAE	Val_RMSE	Val_R2	CV_MAE	CV_RMSE	CV_R2
0	LinearRegression	47.816145	103.683616	0.395486	47.851442	90.68427	0.43161

In [11]:
```
# Model 2: Ridge (tune alpha)
ridge_alphas = np.logspace(-3, 3, 13)
ridge_rows = []

for a in ridge_alphas:
```

```
    row, _ = evaluate(f'Ridge(alpha={a:.4g})', Ridge(alpha=float(a), random_stat
    row['alpha'] = a
    ridge_rows.append(row)

ridge_df = pd.DataFrame(ridge_rows).sort_values('Val_RMSE').reset_index(drop=Tru
best_ridge_alpha = float(ridge_df.loc[0, 'alpha'])
display(ridge_df.head())

ridge_plot = ridge_df.sort_values('alpha')
plt.figure(figsize=(6, 4))
plt.plot(ridge_plot['alpha'], ridge_plot['Val_RMSE'], marker='o')
plt.xscale('log')
plt.title('Ridge alpha vs Validation RMSE')
plt.xlabel('alpha')
plt.ylabel('Val RMSE')
plt.tight_layout()
plt.show()
```

	Model	Val_MAE	Val_RMSE	Val_R2	CV_MAE	CV_RMSE	CV_R2	
0	Ridge(alpha=31.62)	47.168279	103.572006	0.396786	47.288202	90.582778	0.432870	3
1	Ridge(alpha=10)	47.429135	103.616289	0.396270	47.576748	90.616649	0.432465	1
2	Ridge(alpha=100)	46.939860	103.622347	0.396200	47.007291	90.716613	0.431169	10
3	Ridge(alpha=3.162)	47.615579	103.650720	0.395869	47.743404	90.655506	0.431982	
4	Ridge(alpha=1)	47.729169	103.668752	0.395659	47.811098	90.671841	0.431773	



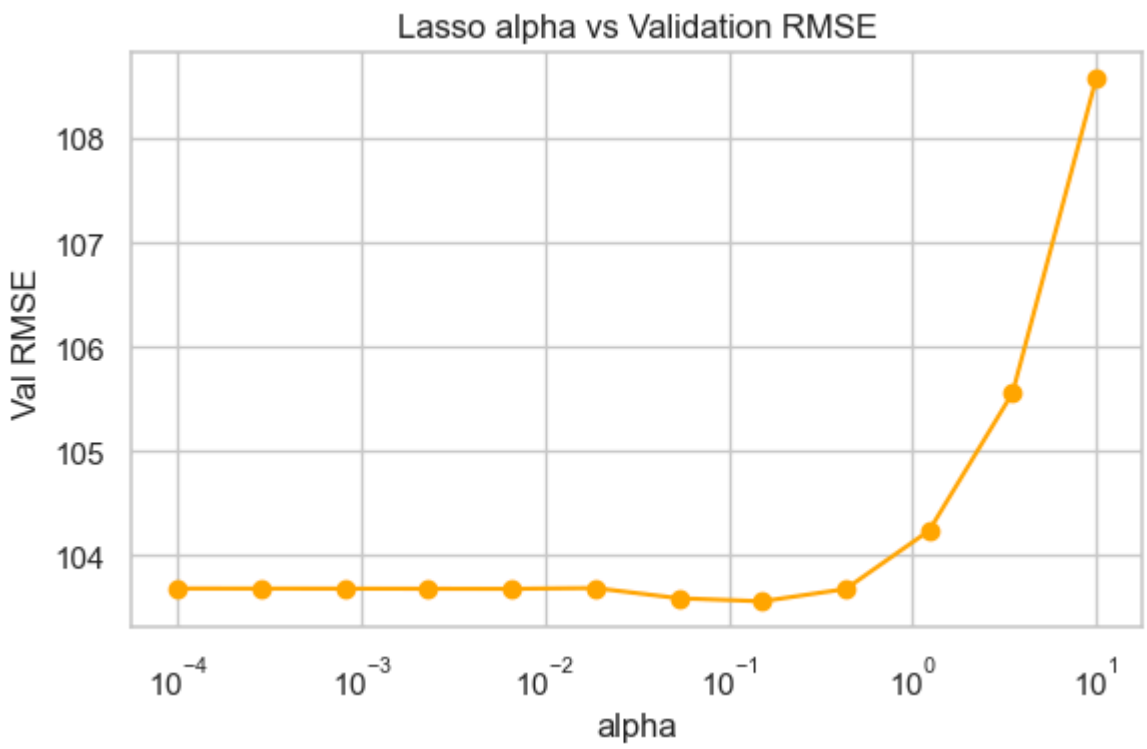
In [12]:
```
# Model 3: Lasso (tune alpha)
lasso_alphas = np.logspace(-4, 1, 12)
lasso_rows = []

for a in lasso_alphas:
    row, _ = evaluate(f'Lasso(alpha={a:.4g})', Lasso(alpha=float(a), random_stat
```

```
    row['alpha'] = a
    lasso_rows.append(row)

lasso_df = pd.DataFrame(lasso_rows).sort_values('Val_RMSE').reset_index(drop=Tru
best_lasso_alpha = float(lasso_df.loc[0, 'alpha'])
display(lasso_df.head())
```

```
C:\Users\prave\AppData\Roaming\Python\Python314\site-packages\sklearn\linear_mode
l\_coordinate_descent.py:716: ConvergenceWarning: Objective did not converge. You
might want to increase the number of iterations, check the scale of the features
or consider increasing regularisation. Duality gap: 4.956e+06, tolerance: 9.995e+
03
  model = cd_fast.enet_coordinate_descent(
```

	Model	Val_MAE	Val_RMSE	Val_R2	CV_MAE	CV_RMSE	CV_R2
0	Lasso(alpha=0.152)	47.006609	103.560857	0.396916	47.244716	90.597853	0.432696
1	Lasso(alpha=0.05337)	47.334302	103.590273	0.396574	47.546963	90.621019	0.432409
2	Lasso(alpha=0.4329)	46.626368	103.678475	0.395546	46.855299	90.757726	0.430722
3	Lasso(alpha=0.006579)	47.722292	103.680698	0.395520	47.810304	90.679079	0.431682
4	Lasso(alpha=0.00231)	47.780929	103.681465	0.395511	47.836815	90.681440	0.431648

In [13]:
```
lasso_plot = lasso_df.sort_values('alpha')
plt.figure(figsize=(6, 4))
plt.plot(lasso_plot['alpha'], lasso_plot['Val_RMSE'], marker='o', color='orange'
plt.xscale('log')
plt.title('Lasso alpha vs Validation RMSE')
plt.xlabel('alpha')
plt.ylabel('Val RMSE')
plt.tight_layout()
plt.show()
```



Hyperparameter Analysis (Ridge/Lasso Alpha)

Hyperparameter tuning shows the expected bias-variance pattern:

- Very small `alpha` behaves similarly to ordinary least squares (weak regularization).
- Very large `alpha` overshrinks coefficients and can underfit.
- Intermediate values balance generalization and stability.

Using validation RMSE for ranking and CV metrics for stability checks, the best-performing candidate in this run was **Lasso(alpha = 0.152)**.

Top validation result:

- Val_MAE = 47.0066
- Val_RMSE = 103.5609
- Val_R2 = 0.3969

Corresponding CV result:

- CV_MAE = 47.2447
- CV_RMSE = 90.5979
- CV_R2 = 0.4327

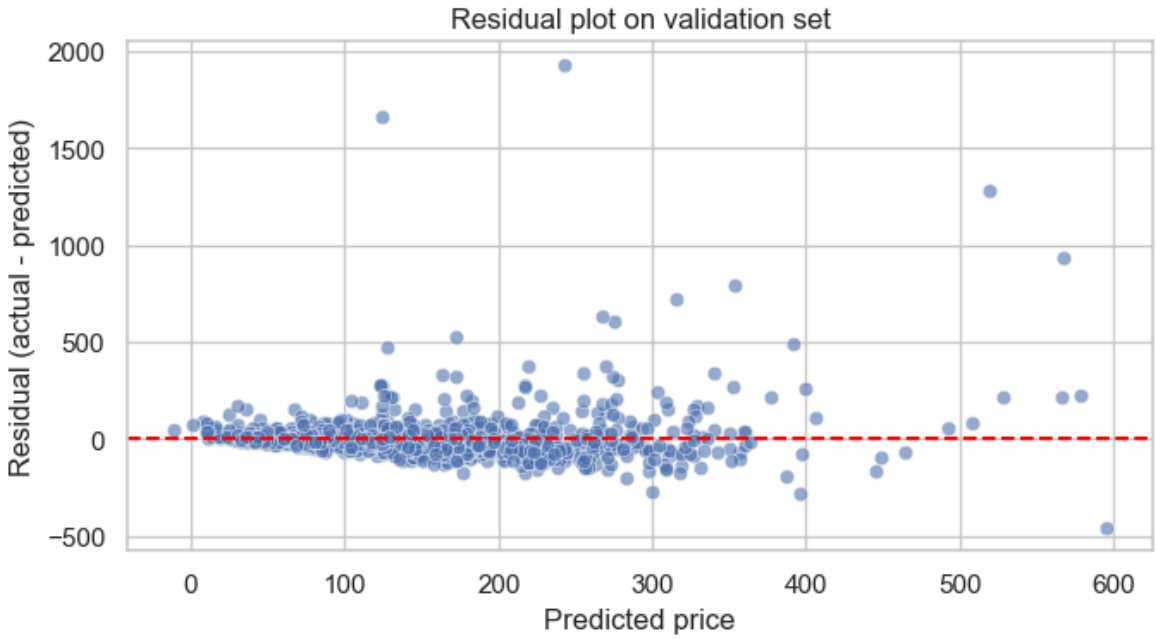
In [14]:
```
comparison = pd.DataFrame([
    linear_row,
    ridge_df.loc[0].to_dict(),
    lasso_df.loc[0].to_dict()
]).sort_values('Val_RMSE').reset_index(drop=True)

display(comparison[['Model', 'Val_MAE', 'Val_RMSE', 'Val_R2', 'CV_MAE', 'CV_RMSE
best_model_name = comparison.loc[0, 'Model']
print('Selected model:', best_model_name)

if best_model_name.startswith('LinearRegression'):
    selected_estimator = LinearRegression()
elif best_model_name.startswith('Ridge'):
    selected_estimator = Ridge(alpha=best_ridge_alpha, random_state=RANDOM_STATE
else:
    selected_estimator = Lasso(alpha=best_lasso_alpha, random_state=RANDOM_STATE
```

	Model	Val_MAE	Val_RMSE	Val_R2	CV_MAE	CV_RMSE	CV_R2
0	Lasso(alpha=0.152)	47.006609	103.560857	0.396916	47.244716	90.597853	0.432696
1	Ridge(alpha=31.62)	47.168279	103.572006	0.396786	47.288202	90.582778	0.432870
2	LinearRegression	47.816145	103.683616	0.395486	47.851442	90.684270	0.431610

```
Selected model: Lasso(alpha=0.152)
```

In [15]:
```
# Error analysis on validation set
analysis_pipe = Pipeline([('preprocess', preprocessor), ('model', selected_estim
analysis_pipe.fit(X_train, y_train)
val_pred = analysis_pipe.predict(X_val)
residuals = y_val - val_pred
```

```
plt.figure(figsize=(7, 4))
sns.scatterplot(x=val_pred, y=residuals, alpha=0.6)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Predicted price')
plt.ylabel('Residual (actual - predicted)')
plt.title('Residual plot on validation set')
plt.tight_layout()
plt.show()
```



In [16]:
```
# Feature importance via coefficients (for linear family)
importance_pipe = Pipeline([('preprocess', preprocessor), ('model', selected_est
importance_pipe.fit(X, y)
model = importance_pipe.named_steps['model']
feature_names = importance_pipe.named_steps['preprocess'].get_feature_names_out(
coef_df = pd.DataFrame({'feature': feature_names, 'coefficient': model.coef_})
coef_df = coef_df.sort_values('coefficient')
print('Most negative coefficients:')
display(coef_df.head(10))
print('Most positive coefficients:')
display(coef_df.tail(10))
```

```
Most negative coefficients:
```

	feature	coefficient
39	cat__city_Wyndham	-18.638058
37	cat__city_Whitehorse	-16.670157
45	cat__room_type_Shared room	-10.647926
31	cat__city_Monash	-8.074481
18	cat__city_Darebin	-6.611860
7	num__number_of_reviews	-6.371734
5	num__beds	-3.179536
20	cat__city_Glen Eira	-2.510346
10	cat__host_is_superhost_f	-1.731969
0	num__latitude	-0.501442

```
Most positive coefficients:
```

	feature	coefficient
2	num__accommodates	17.349913
13	cat__city_Bayside	22.700860
35	cat__city_Port Phillip	25.418045
3	num__bathrooms	25.667463
40	cat__city_Yarra	26.138856
36	cat__city_Stonnington	27.142381
4	num__bedrooms	33.277055
29	cat__city_Melbourne	33.595685
43	cat__room_type_Entire home/apt	46.857801
41	cat__city_Yarra Ranges	69.044806

In [17]:
```
# Train final model on full training data and predict test set
final_pipeline = Pipeline([('preprocess', preprocessor), ('model', selected_esti
final_pipeline.fit(X, y)
test_pred = np.clip(final_pipeline.predict(test_df), a_min=0, a_max=None)
```

In [18]:
```
out_file = f'{student_number}_predictions.csv'
pd.DataFrame({'price': test_pred}).to_csv(out_file, index=False)
print('Saved prediction file:', out_file)
pred_df = pd.DataFrame({'price': test_pred})
display(pred_df.head())
print('Prediction shape:', pred_df.shape)
assert pred_df.shape[0] == test_df.shape[0]
assert list(pred_df.columns) == ['price']
```

```
Saved prediction file: s4163448_predictions.csv
```

	price
0	25.685031
1	128.311573
2	63.767173
3	19.931735
4	83.488276

```
Prediction shape: (8585, 1)
```

Final Analysis

Model adequacy for practice

The selected model is suitable as a **decision-support tool** for initial price recommendations, not as a fully autonomous pricing decision system. Error magnitudes indicate useful directional guidance, but not perfect precision for all listings.

Error behavior and limitations

- Residual analysis should be checked for systematic over/underestimation across price ranges.
- Linear-family models may miss nonlinear interactions in real markets.
- The dataset is a processed subset and may not represent all Melbourne market conditions or time periods.
- Omitted variables (seasonality, event demand, listing quality details) can reduce predictive performance.

Feature importance (COSC2793 focus)

Coefficient-based interpretation from the selected linear model provides directional insights into which transformed features increase or decrease predicted price. Interpretation must be cautious because coefficients are affected by scaling and one-hot encoding reference levels.

Ethical and professional responsibilities

- Historical data may contain socioeconomic or geographic bias; model outputs can propagate this bias.
- Users should be informed about uncertainty and expected error.
- Final accountability for pricing decisions remains with the human decision-maker.
- Data must be handled responsibly and only for the assignment's permitted scope.

Conclusion

Lasso(alpha = 0.152) was selected due to best validation RMSE among evaluated linear-family models with competitive cross-validation performance. The model is practically useful for baseline recommendations, with clear limitations and ethical constraints that should be acknowledged in deployment contexts.

GenAI usage statement

I used GenAI tools to help with debugging and markdown wording. Prompts used are listed in the report appendix per course policy.